Neural Effect Modifier Search

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Abstract

Characterizing heterogeneous treatment effects is essential for understanding *who benefits or is harmed* by an intervention. The task is already challenging with structured and interpretable pre-treatment covariates, i.e., candidate effect modifiers, and becomes even harder when such observations are high-dimensional and entangled (e.g., images or text), requiring interpretation via handcrafted rules or foundation models. Indeed, in an entangled representation, many coordinates may redundantly capture the same underlying source of effect heterogeneity and act as redundant proxy effect modifiers. We focus on this setting in randomized controlled trials with a *known* outcome and, to address this redundancy problem, propose *neural effect modifier search* (NEMS), a recursive conditional-testing procedure that iteratively identifies sufficient and minimal effect-modifier coordinates in an entangled neural representation by conditioning on those already selected. We theoretically prove NEMS consistency and validate it on preliminary synthetic experiments.

1 Introduction

Randomized controlled trials (RCTs) are the canonical design for identifying average causal effects when experimentation is feasible. In many applications, however, the key question is not only whether an intervention works on average, but also how its effect varies across units: "for whom does an intervention help, and for whom does it harm?". Indeed, even when the average treatment effect is close to zero, the corresponding intervention may still be beneficial for one subgroup and harmful for another, e.g., patients with a specific biological profile. A central goal is therefore to identify the *effect modifiers*, namely pre-treatment variables that induce variation in the treatment effect across subpopulations (Rothman et al., 1980; Gail and Simon, 1985).

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Even when candidate effect modifiers are structured and semantically distinct concepts, identifying the primitive drivers of such heterogeneity is statistically delicate due to multiple testing (Bonferroni, 1935) and the distinction between causal effect modification and regression interaction (VanderWeele, 2009). The challenge becomes even sharper in modern settings, where candidate modifiers are often not available as low-dimensional, interpretable covariates. Instead, raw observations are high-dimensional and unstructured, such as radiology images, pathology slides, clinical notes, or gene expression profiles. A common pipeline is therefore to map such raw records into a learned (or rule-based) representation and then test representation coordinates for effect modification (Athey and Imbens, 2016; Athey et al., 2019; Bargagli-Stoffi et al., 2020).

Example (*Clinical Research*).
A randomized trial evaluates the effect of a drug on 30-day mortality in hospitalized patients with severe pneumonia. The trial collects raw chest CT scans and clinical notes, but no curated biomarker panel. The clinicians aim to identify latent signatures, i.e., biomarkers, that characterize treatment-effect heterogeneity by (i) extracting representations from the CT scans and notes, and (ii) selecting the coordinates that capture distinct heterogeneity drivers, e.g., features characterizing subgroups that benefit more or are harmed.

This setting introduces a distinct representation-level difficulty. When the representation *entangles* latent factors, many coordinates can exhibit apparent effect modification simultaneously because they carry overlapping information about the same underlying source of heterogeneity. As experimental power increases, all such proxy coordinates are almost surely detected as effect modifiers, obscuring scientific interpretation by redundancy (Kiernan and Baiocchi, 2022).

In this paper, we study this problem in the *known-outcome* setting, where the outcome of interest is pre-specified and the goal is to identify which latent factors modify the treatment effect. This differs from the *unknown-outcome* exploratory mechanistic setting, which additionally searches for treatment effects in high-dimensional post-treatment measurements (Mencattini et al., 2025) and is outside the scope of this work.

Contributions. We formalize effect modifiers identification from RCTs with sufficient high-dimensional and unstructured pre-treatment observations, together with the aforementioned representation-level redundancy fallacy. We then propose *Neural Effect Modifier Search* (NEMS), a recursive conditional-testing procedure that selects a minimal and sufficient set of representation-level coordinates to characterize HTE while ignoring redundant discoveries by conditioning on those already selected. Assuming principal-alignment and sufficiency, we establish NEMS consistency, analogous in spirit to recursive Neural Effect Search (NES) in Mencattini et al. (2025). Finally, we validate NEMS empirically on synthetic experiments and plan to extend this evaluation to real-world data in future work.

2 Effect modifier identification from high-dimensional and unstructured pre-treatment observations

We consider an i.i.d. sample of units $i \in \mathcal{I} = \{1, \ldots, n\}$. For each unit, we observe $(\boldsymbol{X}_i, T_i, Y_i)$, where $\boldsymbol{X}_i \in \mathcal{X} \subseteq \mathbb{R}^p$ is a high-dimensional pre-treatment observation (e.g., an image or a text record), $T_i \in \{0, 1\}$ is a randomized treatment, and $Y_i \in \mathbb{R}$ is the outcome of interest. Under the potential-outcomes framework (Rubin, 1974), each unit has potential outcomes $(Y_i(1), Y_i(0))$, unit-level treatment effect $\tau_i := Y_i(1) - Y_i(0)$, and average treatment effect $\tau := \mathbb{E}[\tau_i]$.

Let $\boldsymbol{W}_i \in \mathbb{R}^r$ be the (latent) *direct effect modifiers* (VanderWeele and Robins, 2007) encoded in the measurement $\boldsymbol{X}_i$, providing a minimal characterization of treatment-effect heterogeneity. Figure 1 illustrates the corresponding causal model. However, the graph itself does *not* encode effect modification, which we formalize following Hines et al. (2022) and Shin et al. (2025).

Definition 2.1 (Effect modifier)**.** *Fix $j \in \{1, \ldots, r\}$. We say that W_i^j is an* effect modifier *if*

$$\mathbb{P}\Big(\mathbb{E}[\tau_i \mid W_i^j] \neq \tau\Big) > 0. \tag{1}$$

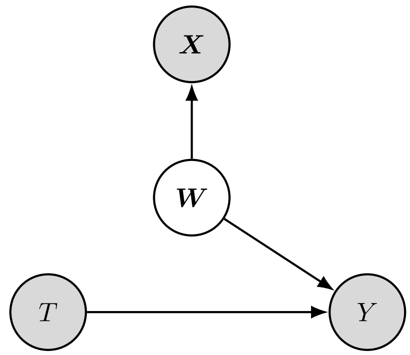

Figure 1: Causal model: in grey the observed variables (raw pre-treatment observation $\boldsymbol{X}$, randomized treatment T, and outcome Y); in white the unobserved ones (latent effect-modification factors $\boldsymbol{W}$).

Definition 2.2 (Conditional effect modifier)**.** *Fix $j \in \{1, \ldots, r\}$ and $S \subseteq \{1, \ldots, r\} \setminus \{j\}$. Let $\boldsymbol{W}_i^S = \{W_i^k\}_{k \in S}$. We say that W_i^j is a* conditional effect modifier relative to *S if*

$$\mathbb{P}\Big(\mathbb{E}[\tau_i \mid W_i^j, \boldsymbol{W}_i^S] \neq \mathbb{E}[\tau_i \mid \boldsymbol{W}_i^S]\Big) > 0. \tag{2}$$

For each individual we assume:

A1 (Randomization and SUTVA).

$$T_i \perp (Y_i(0), Y_i(1), \boldsymbol{W}_i, \boldsymbol{X}_i), \tag{3}$$

$$Y_i = Y_i(T_i) = Y_i(T_1, \ldots, T_i, \ldots, T_n). \tag{4}$$

to identify treatment-effect heterogeneity, and

A2 (Observation sufficiency). There exists a measurable map $g : \mathcal{X} \to \mathbb{R}^r$ such that $\boldsymbol{W}_i = g(\boldsymbol{X}_i)$ and

$$\mathbb{E}[\tau_i \mid \boldsymbol{X}_i] \overset{a.s.}{=} \mathbb{E}[\tau_i \mid \boldsymbol{W}_i]. \tag{5}$$

so the relevant effect modifiers are measurable in the raw observations.

We then extract the representations

$$\boldsymbol{Z}_i = \phi(\boldsymbol{X}_i) \in \mathbb{R}^m, \tag{6}$$

whose coordinates[1] $Z_i^1, \ldots, Z_i^m$ are the candidate neural effect modifier. Here ϕ is a (pretrained) representation map, possibly followed by an interpretable feature extractor (e.g., an SAE), or a hand-crafted feature map. Note that $\boldsymbol{Z}_i$ is a higher-level representation of $\boldsymbol{X}_i$: it is typically richer than $\boldsymbol{W}_i$ ($m \gg r$) and may mix information about the same component of $\boldsymbol{W}_i$, i.e.,

[1] Equivalently, "*neurons*".

it may exhibit *artificial* entanglement. In addition, we assume:

A3 (Principal alignment).

$$\boldsymbol{Z}_i = A\boldsymbol{W}_i + \boldsymbol{\xi}_i, \tag{7}$$

where $\boldsymbol{\xi}_i \perp (T_i, \boldsymbol{W}_i)$ and $\mathbb{E}[\boldsymbol{\xi}_i] = \mathbf{0}$. For each $k \in \{1, \dots, r\}$, there exists $j_k \in \{1, \dots, m\}$ such that

$$|A_{j_k k}| \geq |A_{jk}| + \Delta \qquad \forall j \neq j_k \tag{8}$$

for some $\Delta > 0$, and W_i^k is a conditional effect modifier relative to $\{1, \dots, r\} \setminus \{k\}$.

so the representation contains one sufficiently aligned coordinate for each relevant heterogeneity factor, and

A4 (Valid and consistent p-values). For each $j \in \{1, \dots, m\}$ and $S \subseteq \{1, \dots, m\} \setminus \{j\}$, let $p_j(S)$ denote the p-value of the test for

$$\mathbb{E}[\tau_i \mid Z_i^j, \boldsymbol{Z}_i^S] \overset{a.s.}{=} \mathbb{E}[\tau_i \mid \boldsymbol{Z}_i^S]. \tag{9}$$

If Equation 9 holds, then

$$\mathbb{P}\big(p_j(S) \leq \alpha\big) \leq \alpha \qquad \forall \alpha \in [0, 1]. \tag{10}$$

If Equation 9 fails under a fixed alternative, then

$$p_j(S) \xrightarrow{p} 0 \qquad \text{as } n \to \infty. \tag{11}$$

so the tests used below are valid under the null and asymptotically detect any fixed non-null signal.

Task. Given $\{(\boldsymbol{X}_i, T_i, Y_i)\}_{i=1}^n$, extract interpretable representations $\boldsymbol{Z}_i$ from $\boldsymbol{X}_i$, and select coordinates $\mathcal{J}^* \subseteq \{1, \dots, m\}$ containing one representative coordinate for each latent *direct* effect modifier encoded in $\boldsymbol{W}_i$.

3 Paradox of marginal testing for neural effect modifiers

The main challenge is not missing true signals, but distinguishing direct effect modifiers from redundant proxies. In entangled representations, several coordinates may exhibit effect-modification signal because they encode overlapping information about the same latent factor. Given $\boldsymbol{Z}_i$, define the set of neural effect modifier

$$\mathcal{J} \coloneqq \Big\{ j \in \{1, \dots, m\} : \mathbb{E}[\tau_i \mid Z_i^j] \overset{a.s.}{\neq} \tau \Big\}. \tag{12}$$

Proposition 3.1 (Marginal tests detect every marginal signal)**.** *Assume A1, A2, and A4 with $S = \emptyset$. Fix $j \in \mathcal{J}$. Then, for any fixed $\alpha \in (0, 1)$,*

$$\mathbb{P}\big(p_j(\emptyset) \leq \alpha\big) \to 1 \qquad \text{as } n \to \infty. \tag{13}$$

Proof sketch. Since $j \in \mathcal{J}$, Equation 12 implies that Equation 9 fails with $S = \emptyset$. Hence, by Equation 11, $p_j(\emptyset) \xrightarrow{p} 0$, which yields Equation 13.

Interpretation. Proposition 3.1 shows that increasing experimental power eventually reveals every coordinate carrying marginal effect-modification signal. The problem is therefore not lack of detection, but lack of minimality. Under A3, the same latent effect modifier can be encoded by several coordinates, so more power may recover more redundant proxy coordinates rather than a smaller and more interpretable set.

4 Neural Effect Modifier Search

To recover a minimal representative set rather than all marginal proxies, we propose *Neural Effect Modifier Search* (NEMS), a greedy recursive conditional-testing procedure. For $j \in [m]$ and $S \subseteq [m] \setminus \{j\}$, define

$$H_0(j \mid S): \quad \mathbb{E}[\tau_i \mid Z_i^j, \boldsymbol{Z}_i^S] \overset{a.s.}{=} \mathbb{E}[\tau_i \mid \boldsymbol{Z}_i^S]. \tag{14}$$

Under A1, this can be tested via a treatment–interaction model such as

$$\begin{aligned} Y_i = {} & \beta_0 + \beta_T T_i + \beta_j Z_i^j + \beta_S^\top \boldsymbol{Z}_i^S \\ & + \gamma_j (T_i Z_i^j) + \gamma_S^\top (T_i \boldsymbol{Z}_i^S) + \varepsilon_i, \end{aligned} \tag{15}$$

testing $H_0 : \gamma_j = 0$ with either a randomization test or an asymptotic Wald/t-test (Imbens and Rubin, 2015; Gao, 2025).

Starting from $S = \emptyset$, NEMS computes $p_j(S)$ over the remaining coordinates, selects the smallest one, and adds it only if it passes a Bonferroni gate over the remaining hypotheses. It proceeds until the gate is blocked and then the selected coordinates are interpreted, e.g., top-activating examples, concept probes, or summaries, to map them back to meaningful pretreatment factors.

In practice, NEMS is run at a fixed level α, as in our experiments. The next result is a stronger asymptotic refinement: under a vanishing schedule $\alpha_n \downarrow 0$ and two (additional) proof-only recursive regularity conditions stated in Appendix B, the same recursion achieves exact support recovery.

Algorithm 1 neural effect modifier Search (NEMS)

Input: $\{(\boldsymbol{X}_i, T_i, Y_i)\}_{i=1}^n$, representation map ϕ, level α
Compute $\boldsymbol{Z}_i \leftarrow \phi(\boldsymbol{X}_i) \in \mathbb{R}^m$ for all i
Initialize $S \leftarrow \emptyset$
while true **do**
  $M \leftarrow [m] \setminus S$
  for $j \in M$ **do**
    Compute $p_j(S)$ for Equation 14
  end for
  $j^\star \leftarrow \arg\min_{j \in M} p_j(S)$
  if $p_{j^\star}(S) > \alpha/|M|$ **then**
    break
  else
    $S \leftarrow S \cup \{j^\star\}$
  end if
end while
Output: S

Theorem 4.1 (Exact recovery under a vanishing level schedule)**.** *Assume A1–A4. Under the additional proof-only recursive regularity conditions stated in Appendix B, if Algorithm 1 is analyzed with $\alpha = \alpha_n \downarrow 0$, then, for fixed m,*

$$\mathbb{P}\big(S_{\mathrm{NEMS}}^{(n)} = \mathcal{J}^*\big) \to 1 \qquad \text{as } n \to \infty, \tag{16}$$

where $S_{\mathrm{NEMS}}^{(n)}$ is the output of Algorithm 1 at sample size n.

Proof sketch. The appendix shows that, after conditioning on any $S \subseteq \mathcal{J}^*$, the only remaining non-null coordinates are $\mathcal{J}^* \setminus S$, and they still cross the shrinking Bonferroni threshold with probability tending to one. Hence NEMS adds one new representative at each step, never adds a null coordinate, and stops exactly at $\mathcal{J}^*$. With fixed α, the same argument yields only a weaker conservative lower bound. See Appendix B.

5 Synthetic Experiments

We evaluate the statistical selection behavior of NEMS independently of representation learning using a controlled synthetic benchmark. We simulate a randomized binary treatment T, two latent *direct* effect modifiers $\boldsymbol{W} = (W_1, W_2)$, a continuous outcome Y, and an entangled representation $\boldsymbol{Z}$ satisfying A3 (see Appendix C for full data-generating details). We repeat the analysis under both linear and quadratic heterogeneity, and report here the former, comparing NEMS (ours) with Marginal-test and Marginal-test with Bonferroni adjustment (Bonferroni, 1935). We evaluate selection using intersection-over-union (IoU) (with precision and recall reported in the appendix), varying sample size and effect size, and repeating each configuration over 50 random seeds.

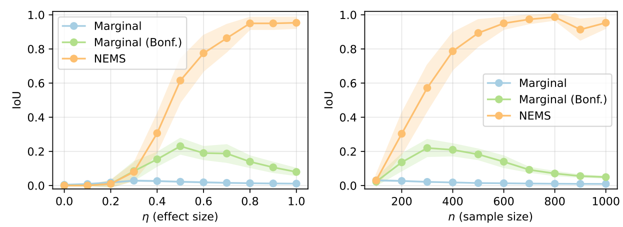

Figure 2: Synthetic experiments with linear heterogeneity, varying effect size and sample size (see Appendix C for full experimental details), reporting IoU with 95% confidence intervals from 50 repetitions per configuration.

Results are reported in Figure 2, confirming the theoretical finding: NEMS consistently achieves higher IoU as experimental power increases (either through larger effect size or larger sample size), up to perfect recovery, while marginal procedures remain limited by redundant discoveries induced by entangled representations, i.e., low precision. Marginal-test with Bonferroni adjustment slightly improves precision, but remains far from isolating only the direct effect modifiers.

6 Conclusion

Effect modifier identification from RCTs becomes substantially harder when candidate modifiers are not curated covariates, but coordinates of an entangled baseline representation. In this setting, representation-level non-uniqueness creates redundant discoveries under standard effect-modification definitions, so naive marginal screening may fail to provide a minimal, non-redundant, and interpretable characterization of heterogeneity. NEMS addresses this issue through recursive conditional testing, aiming to recover representation coordinates that are sufficient yet minimal, i.e., non-redundant, for characterizing treatment-effect heterogeneity.

A current limitation is that our preliminary validation isolates the statistical selection problem using synthetic representations. Future work will evaluate the full pipeline, including the representation-learning component, on semi-synthetic and real datasets, and will further examine the practical validity of the identifying assumptions. More broadly, we hope this work helps establish effect-modifier identification in neural representations as a distinct and tractable statistical problem, and provides a useful first step toward more interpretable heterogeneity analysis from high-dimensional pre-treatment data.

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Supplementary Materials

A Related Work

Representation learning for causal downstream tasks. A growing line of work studies how high-dimensional and unstructured observations can be converted into lower-dimensional variables for downstream causal analysis (Cadei et al., 2024, 2025; Yao et al., 2025a,b; Mencattini et al., 2025). In practice, raw modalities such as images or text are often mapped into learned or hand-crafted representations, whose coordinates are then used as proxy covariates in estimation or testing pipelines (Cadei et al., 2024, 2025; Yao et al., 2025a,b). Our setting fits this broader paradigm: a representation map ϕ turns the raw pre-treatment observation $\boldsymbol{X}$ into candidate coordinates $\boldsymbol{Z}$, but the main challenge is not learning ϕ itself. Rather, once such a representation is available, the question is how to perform effect-modifier identification over its coordinates in a statistically meaningful and interpretable way. In this sense, NEMS is complementary to representation-learning approaches: it acts as a downstream causal selection layer over neural features, aiming to isolate a minimal set of coordinates that captures treatment-effect heterogeneity rather than to recover the representation itself.

Heterogeneous treatment effect estimation and effect modifier identification. The classical literature on heterogeneous treatment effects studies how treatment responses vary across structured pre-treatment covariates, often using trees, forests, or rule-based procedures to identify subgroups with different effects (Athey and Imbens, 2016; Athey et al., 2019; Bargagli-Stoffi et al., 2020). Related recent work has sharpened the statistical notion of effect modification itself, distinguishing variable-level effect modifiers from generic predictive interactions and formalizing conditional notions of treatment-effect relevance (Hines et al., 2022; Shin et al., 2025; VanderWeele, 2009). Our work is aligned with this objective, but differs in the regime it targets: here, candidate modifiers are not curated low-dimensional covariates, but coordinates of an entangled neural representation extracted from high-dimensional pre-treatment measurements. Consequently, the main difficulty is no longer only flexible HTE estimation, but identifying a *minimal and sufficient* subset of representation coordinates. In particular, many coordinates may act as redundant proxy effect modifiers because they carry overlapping information about the same latent heterogeneity factor (VanderWeele and Robins, 2007), a failure mode not addressed by standard marginal subgroup-discovery procedures.

Exploratory causal inference. Closest in spirit is *Exploratory Causal Inference in SAEnce*, which introduced Neural Effect Search (NES) for the complementary *unknown-outcome* setting (Mencattini et al., 2025). There, the goal is to discover which latent outcome coordinates in a high-dimensional *post-treatment* measurement are affected by the intervention. Here, instead, the outcome Y is known in advance and the objective is to identify which latent *pre-treatment* factors modify its treatment effect. NEMS inherits the same high-level recursive-testing intuition as NES, but the causal target is different: NES asks *what is affected*, while NEMS asks *for whom the effect differs*. This shift removes the exploratory outcome-discovery component and turns the core statistical problem into one of pruning redundant proxy modifiers in baseline neural representations. Accordingly, NEMS can be viewed as the known-outcome, effect-modifier counterpart of NES (Mencattini et al., 2025).

B Proofs

The main text defines NEMS for a generic level α, which is the practical version used in our experiments. This appendix proves a stronger asymptotic refinement: when the same recursion is analyzed under a vanishing schedule $\alpha_n \downarrow 0$, exact support recovery follows under two proof-only recursive regularity conditions.

Let

$$\mathcal{J}^* := \{j_1, \dots, j_r\} \subseteq [m] \tag{17}$$

denote the representative coordinate set induced by Assumption A3, and define

$$M(S) \coloneqq [m] \setminus S \qquad \text{for any } S \subseteq [m]. \tag{18}$$

Proof-only recursive regularity conditions. We assume, only in this appendix, that:

(i) for every $S \subseteq \mathcal{J}^*$ and every $j \in M(S)$,

$$\mathbb{P}\Big(\mathbb{E}[\tau_i \mid Z_i^j, \boldsymbol{Z}_i^S] \neq \mathbb{E}[\tau_i \mid \boldsymbol{Z}_i^S]\Big) > 0 \quad \Longleftrightarrow \quad j \in \mathcal{J}^* \setminus S; \tag{19}$$

(ii) the nominal level satisfies

$$\alpha_n \downarrow 0, \tag{20}$$

and for every $S \subseteq \mathcal{J}^*$ and $j \in \mathcal{J}^* \setminus S$,

$$\mathbb{P}\left(p_j(S) \leq \frac{\alpha_n}{|M(S)|}\right) \to 1. \tag{21}$$

B.1 Proof of Proposition 3.1

Proposition. Assume A1, A2, and A4 with $S = \emptyset$. Define

$$\mathcal{J} \coloneqq \left\{j \in [m] : \mathbb{E}[\tau_i \mid Z_i^j] \overset{a.s.}{\neq} \tau\right\}. \tag{22}$$

Then, for any $j \in \mathcal{J}$ and any fixed $\alpha \in (0, 1)$,

$$\mathbb{P}\big(p_j(\emptyset) \leq \alpha\big) \to 1 \qquad \text{as } n \to \infty. \tag{23}$$

Proof. Fix $j \in \mathcal{J}$. By Equation 22,

$$\mathbb{E}[\tau_i \mid Z_i^j] \overset{a.s.}{\neq} \tau = \mathbb{E}[\tau_i]. \tag{24}$$

Hence the null in Equation 9 fails when $S = \emptyset$. By Equation 11,

$$p_j(\emptyset) \xrightarrow{p} 0. \tag{25}$$

Therefore, for every fixed $\alpha \in (0, 1)$,

$$\mathbb{P}\big(p_j(\emptyset) \leq \alpha\big) \to 1, \tag{26}$$

which proves Equation 23. □

B.2 Proof of Theorem 4.1

Theorem. Assume A1–A4 and the proof-only conditions in Equations 19–21. Let $S_{\text{NEMS}}^{(n)}$ be the output of Algorithm 1 at sample size n, analyzed with $\alpha = \alpha_n$. Then, for fixed m,

$$\mathbb{P}\big(S_{\text{NEMS}}^{(n)} = \mathcal{J}^*\big) \to 1 \qquad \text{as } n \to \infty. \tag{27}$$

More precisely,

$$\mathbb{P}\big(S_{\text{NEMS}}^{(n)} = \mathcal{J}^*\big) \geq 1 - 2^r \alpha_n - o(1). \tag{28}$$

Proof. Define

$$E_n \coloneqq \bigcap_{S \subseteq \mathcal{J}^*} \bigcap_{j \in \mathcal{J}^* \setminus S} \left\{p_j(S) \leq \frac{\alpha_n}{|M(S)|}\right\}, \tag{29}$$

and

$$G_n \coloneqq \bigcap_{S \subseteq \mathcal{J}^*} \bigcap_{j \in [m] \setminus \mathcal{J}^*} \left\{p_j(S) > \frac{\alpha_n}{|M(S)|}\right\}. \tag{30}$$

By Equation 21, every event in the intersection defining E_n has probability tending to one. Since m and r are fixed, the intersection is finite, hence

$$\mathbb{P}(E_n) \to 1. \tag{31}$$

Next fix $S \subseteq \mathcal{J}^*$. By Equation 19, every $j \in [m] \setminus \mathcal{J}^*$ satisfies the null in Equation 9. Therefore, by Equation 10 and a union bound,

$$\mathbb{P}\left(\bigcup_{j \in [m] \setminus \mathcal{J}^*} \left\{ p_j(S) \le \frac{\alpha_n}{|M(S)|} \right\} \right) \le \sum_{j \in [m] \setminus \mathcal{J}^*} \frac{\alpha_n}{|M(S)|}. \tag{32}$$

Since $|[m] \setminus \mathcal{J}^*| = m - r$ and $S \subseteq \mathcal{J}^*$ implies $|M(S)| = m - |S| \ge m - r$,

$$\mathbb{P}\left(\bigcup_{j \in [m] \setminus \mathcal{J}^*} \left\{ p_j(S) \le \frac{\alpha_n}{|M(S)|} \right\} \right) \le \alpha_n. \tag{33}$$

There are 2^r possible subsets $S \subseteq \mathcal{J}^*$, so

$$\mathbb{P}(G_n^c) \le 2^r \alpha_n. \tag{34}$$

We now show that on $E_n \cap G_n$, Algorithm 1 returns exactly $\mathcal{J}^*$. Start from $S = \emptyset$. Suppose at some iteration the current selected set satisfies $S \subseteq \mathcal{J}^*$.

If $S \neq \mathcal{J}^*$, then by E_n, every $j \in \mathcal{J}^* \setminus S$ satisfies

$$p_j(S) \le \frac{\alpha_n}{|M(S)|}, \tag{35}$$

so at least one remaining coordinate passes the Bonferroni gate. By G_n, every $j \in [m] \setminus \mathcal{J}^*$ satisfies

$$p_j(S) > \frac{\alpha_n}{|M(S)|}, \tag{36}$$

so no non-representative coordinate passes. Hence the minimizer selected by Algorithm 1 must belong to $\mathcal{J}^* \setminus S$, and the next selected set remains a subset of $\mathcal{J}^*$.

Thus, on $E_n \cap G_n$, the recursion adds exactly one new representative at each successful step and never adds a non-representative coordinate. After r steps, it reaches $S = \mathcal{J}^*$. At that point, by G_n, every remaining coordinate fails the Bonferroni gate, so the algorithm stops. Therefore,

$$E_n \cap G_n \subseteq \left\{ S_{\mathrm{NEMS}}^{(n)} = \mathcal{J}^* \right\}. \tag{37}$$

Combining Equations 31 and 34,

$$\mathbb{P}\big(S_{\mathrm{NEMS}}^{(n)} = \mathcal{J}^*\big) \ge 1 - \mathbb{P}(E_n^c) - \mathbb{P}(G_n^c) \ge 1 - 2^r \alpha_n - o(1). \tag{38}$$

This proves Equation 28, and Equation 27 follows since $\alpha_n \downarrow 0$. □

Remark on fixed α. If NEMS is instead run at a fixed level α, the same argument yields only

$$\liminf_{n \to \infty} \mathbb{P}\big(S_{\mathrm{NEMS}}^{(n)} = \mathcal{J}^*\big) \ge 1 - 2^r \alpha. \tag{39}$$

This lower bound is conservative: it follows only from worst-case null validity and does not preclude substantially better empirical behavior. The vanishing schedule $\alpha_n \downarrow 0$ is therefore used only to obtain the stronger exact-recovery asymptotic refinement above.

C Synthetic experiment details

This appendix specifies the synthetic experiments used to evaluate the statistical behavior of NEMS in a controlled setting. The goal is to isolate the *selection* problem (redundant vs. non-redundant modifier coordinates) from representation learning by directly simulating latent heterogeneity factors $\boldsymbol{W}$, randomized treatment T, outcomes Y, and an entangled representation $\boldsymbol{Z} = A\boldsymbol{W} + \boldsymbol{\xi}$.

All experiments use $r = 2$ latent heterogeneity factors and $m = 500$ candidate representation coordinates. By construction, two coordinates are principally aligned with the two latent effect-modifier components (Assumption A3), and these two coordinates define the ground-truth target set for recovery.

C.1 Data-generating process

For each unit $i = 1, \ldots, n$, we generate $\boldsymbol{W}_i$ and T_i as

$$\boldsymbol{W}_i = (W_{i1}, W_{i2})^\top \sim \mathcal{N}(0, I_2), \tag{40}$$

$$T_i \sim \text{Bernoulli}(0.5), \tag{41}$$

independently across units, and independently of $\boldsymbol{W}_i$.

We then generate an entangled representation $\boldsymbol{Z}_i \in \mathbb{R}^m$ as

$$\boldsymbol{Z}_i = A\boldsymbol{W}_i + \boldsymbol{\xi}_i, \tag{42}$$

with

$$\boldsymbol{\xi}_i \sim \mathcal{N}(0, \sigma_\xi^2 I_m), \tag{43}$$

where $m = 500$ and $\sigma_\xi = 1$.

The loading matrix $A \in \mathbb{R}^{500\times 2}$ is constructed to satisfy principal alignment (A3) with known principal coordinates $j_1 = 1$ and $j_2 = 2$. Specifically,

$$A_{1,\cdot} = (a_{\text{main}}, 0), \tag{44}$$

$$A_{2,\cdot} = (0, a_{\text{main}}), \tag{45}$$

with $a_{\text{main}} = 1.5$. For the remaining coordinates $j = 3, \ldots, 500$, we draw provisional loadings

$$\tilde{A}_{jk} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_A^2), \qquad \sigma_A = 0.35, \tag{46}$$

and clip coordinatewise to $[-a_{\max}, a_{\max}]$, with $a_{\max} = 1.0$:

$$A_{jk} = \text{clip}(\tilde{A}_{jk}, -1.0, 1.0), \qquad k \in \{1, 2\}. \tag{47}$$

Hence, for $k \in \{1, 2\}$,

$$|A_{j_k,k}| = 1.5 \geq 1.0 + 0.5 \geq |A_{j,k}| + 0.5 \quad \forall j \neq j_k, \tag{48}$$

so A3 holds with separation margin $\Delta = 0.5$. The remaining 498 coordinates are entangled mixtures of the same latent factors.

Within each replicate, all coordinates of $\boldsymbol{Z}$ are standardized using sample mean and sample standard deviation before testing.

C.2 Potential outcomes and observed outcome

We generate outcomes from a structural form with additive baseline signal and treatment-effect heterogeneity:

$$Y_i(0) = \mu(\boldsymbol{W}_i) + \varepsilon_i^Y, \tag{49}$$

$$Y_i(1) = \mu(\boldsymbol{W}_i) + \tau(\boldsymbol{W}_i) + \varepsilon_i^Y, \tag{50}$$

where

$$\varepsilon_i^Y \sim \mathcal{N}(0, \sigma_Y^2), \qquad \sigma_Y = 1, \tag{51}$$

and the observed outcome is

$$Y_i = Y_i(T_i). \tag{52}$$

The baseline component is

$$\mu(\boldsymbol{W}_i) = \beta_1 W_{i1} + \beta_2 W_{i2} + \beta_{12} W_{i1} W_{i2}, \tag{53}$$

with $(\beta_1, \beta_2, \beta_{12}) = (0.5, -0.5, 0.25)$.

The treatment effect $\tau(\boldsymbol{W}_i)$ is controlled by an effect-size parameter $\eta \geq 0$. We use two variants.

Linear heterogeneity.

$$\tau(\boldsymbol{W}_i) = \tau_0 + \eta\, \boldsymbol{w}^\top \boldsymbol{W}_i, \tag{54}$$

with

$$\boldsymbol{w} = \frac{1}{\sqrt{2}}(1,1)^\top, \qquad \tau_0 = 0. \tag{55}$$

Quadratic heterogeneity.

$$\tau(\boldsymbol{W}_i) = \tau_0 + \eta \left(\boldsymbol{w}^\top \boldsymbol{W}_i + \lambda_q (W_{i1}^2 - W_{i2}^2) \right), \tag{56}$$

with

$$\lambda_q = 0.5, \qquad \tau_0 = 0. \tag{57}$$

The main text reports the linear-heterogeneity experiments. The quadratic variant is used as an appendix ablation under the same protocol.

C.3 Methods and testing procedure

We compare NEMS against two baselines, Marginal-test and Marginal-test with Bonferroni adjustment, at significance level $\alpha = 0.05$.

All methods are based on testing treatment–feature interactions in linear working models with HC3 robust standard errors and two-sided tests. The reported experiments use the continuous-outcome specification above.

For each candidate coordinate j, the marginal interaction model is

$$Y_i = \beta_0 + \beta_T T_i + \beta_j Z_{ij} + \gamma_j (T_i Z_{ij}) + e_i, \tag{58}$$

and we test

$$H_{0,j} : \gamma_j = 0 \quad \text{vs.} \quad H_{1,j} : \gamma_j \neq 0. \tag{59}$$

Let p_j denote the resulting two-sided p-value for the interaction coefficient.

(i) Marginal-test: select

$$\mathtt{S}_{\text{marg}} = \{ j \in [m] : p_j \leq \alpha \}. \tag{60}$$

(ii) Marginal-test with Bonferroni adjustment: select

$$\mathtt{S}_{\text{marg-bonf}} = \{ j \in [m] : p_j \leq \alpha / m \}. \tag{61}$$

(iii) NEMS: recursively perform conditional interaction tests. Starting from $\mathtt{S} = \varnothing$, for each unselected coordinate $j \notin \mathtt{S}$, fit

$$Y_i = \beta_0 + \beta_T T_i + \beta_j Z_{ij} + \boldsymbol{\beta}_{\mathtt{S}}^\top \boldsymbol{Z}_i^{\mathtt{S}} + \gamma_j (T_i Z_{ij}) + \boldsymbol{\gamma}_{\mathtt{S}}^\top (T_i \boldsymbol{Z}_i^{\mathtt{S}}) + e_i, \tag{62}$$

and test

$$H_{0,j|\mathtt{S}} : \gamma_j = 0 \quad \text{vs.} \quad H_{1,j|\mathtt{S}} : \gamma_j \neq 0, \tag{63}$$

obtaining a two-sided HC3-robust p-value $p_j(\mathtt{S})$ for the additional interaction term $T_i Z_{ij}$.

Define

$$\bar{j} = \arg\min_{j \notin \mathtt{S}} p_j(\mathtt{S}). \tag{64}$$

If

$$p_{\bar{j}}(\mathtt{S}) \leq \frac{\alpha}{|[m] \setminus \mathtt{S}|}, \tag{65}$$

add $\bar{j}$ to $\mathtt{S}$; otherwise stop. The final selected set is

$$\mathtt{S}_{\text{NEMS}} = \mathtt{S}. \tag{66}$$

This implements a recursive conditional selection rule with Bonferroni correction over the remaining coordinates at each step.

C.4 Ground truth and evaluation metrics

By construction, the two principally aligned coordinates are

$$\mathcal{J}^* = \{1, 2\}. \tag{67}$$

We treat $\mathcal{J}^*$ as the ground-truth representative set of latent heterogeneity directions.

For a selected set $\mathtt{S} \subseteq [m]$, we report intersection-over-union (IoU),

$$\text{IoU}(\mathtt{S}, \mathcal{J}^*) = \frac{|\mathtt{S} \cap \mathcal{J}^*|}{|\mathtt{S} \cup \mathcal{J}^*|}, \tag{68}$$

which rewards recovery of the principal coordinates while penalizing redundant discoveries.

In the appendix, we additionally report precision and recall. Precision is

$$\text{Precision}(\mathtt{S}, \mathcal{J}^*) = \begin{cases} \frac{|\mathtt{S} \cap \mathcal{J}^*|}{|\mathtt{S}|}, & |\mathtt{S}| > 0, \\ 0, & |\mathtt{S}| = 0, \end{cases} \tag{69}$$

and recall is

$$\text{Recall}(\mathtt{S}, \mathcal{J}^*) = \frac{|\mathtt{S} \cap \mathcal{J}^*|}{|\mathcal{J}^*|} = \frac{|\mathtt{S} \cap \mathcal{J}^*|}{2}. \tag{70}$$

C.5 Experimental protocol

We run two sweeps, each over 50 random seeds. In the reported experiments, the loading matrix A is sampled once and then kept fixed across all seeds and all sweeps. Across seeds, we resample $\boldsymbol{W}$, T, $\boldsymbol{\xi}$, and ε^Y. This keeps the representation geometry (and thus the redundancy structure) fixed while isolating the statistical effect of experimental power. Reported curves show the mean across seeds, with 95% confidence intervals.

Effect-size sweep

We fix $n = 1000$ and vary the heterogeneity strength over

$$\eta \in \{0.0, 0.1, 0.2, \ldots, 1.0\}. \tag{71}$$

Within each seed, we keep the draws of $\boldsymbol{W}$, T, $\boldsymbol{\xi}$, and ε^Y fixed across all η, and only vary $\tau(\boldsymbol{W})$. This reduces Monte Carlo noise and isolates the effect of signal strength.

Sample-size sweep

We fix $\eta = 0.6$ and vary

$$n \in \{100, 200, 300, \ldots, 1000\}. \tag{72}$$

Within each seed, we generate one dataset at $n_{\max} = 1000$ and evaluate smaller sample sizes using nested prefixes (the first n units), which yields smoother comparisons across methods.

C.6 Additional ablations

The main text reports only the two IoU plots for the linear heterogeneity specification: (i) effect size versus IoU and (ii) sample size versus IoU.

In the appendix, we additionally report:

(i) the same two sweeps under the quadratic heterogeneity specification (IoU);

(ii) precision and recall (in addition to IoU) for both linear and quadratic heterogeneity, under the same effect-size and sample-size sweeps.

The empirical results are consistent with the theory across both sweeps (varying effect size and varying sample size) and across both heterogeneity specifications (linear in the main text, quadratic in this appendix).

NEMS generally and often substantially outperforms both baselines in IoU. This holds both when increasing effect size η (stronger heterogeneity signal) and when increasing sample size n (higher experimental power). In particular, even at high power, the marginal procedures do not reliably solve the target selection problem: they can detect many interaction-associated coordinates, but they do not remove redundancy induced by entangled representations, and therefore their IoU remains limited.

The precision–recall decomposition clarifies the baseline trade-off. Marginal-test with Bonferroni adjustment typically improves precision relative to Marginal-test, reflecting stronger control against false discoveries, but this comes at the cost of lower recall, especially at low power. By contrast, Marginal-test often attains relatively high recall even when power is limited, but this is accompanied by poor precision (and thus poor IoU), consistent with selecting many redundant coordinates.

NEMS improves IoU by combining detection with recursive conditional pruning. The conditional tests remove coordinates whose apparent interaction signal is explained by already selected interaction terms, which is precisely the failure mode of marginal screening in entangled representations.

In very low-power regimes, one may also consider an exploratory variant of NEMS that omits (or relaxes) the Bonferroni correction in the recursive step to improve sensitivity. This can be useful when the goal is hypothesis generation rather than controlled confirmation. However, this no longer provides the same family-wise error control and should therefore be reported explicitly as an exploratory procedure.

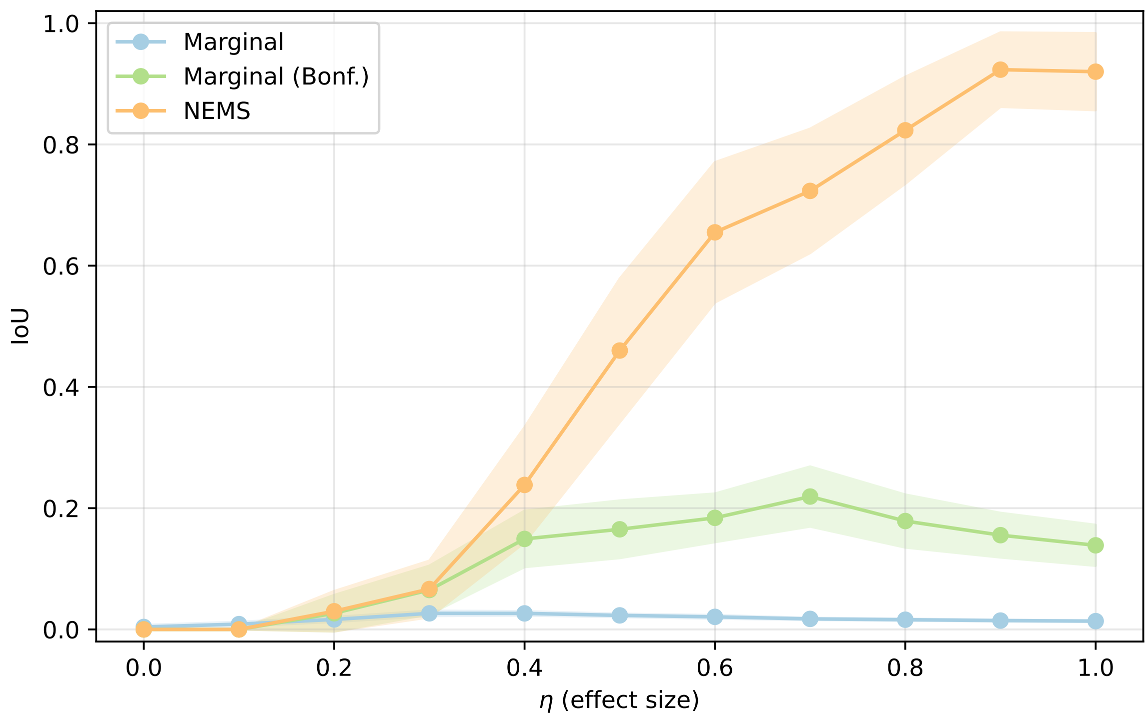

Figure 3: Synthetic experiments with quadratic heterogeneity, varying effect size, reporting IoU with 95% confidence intervals from 50 repetitions per configuration.

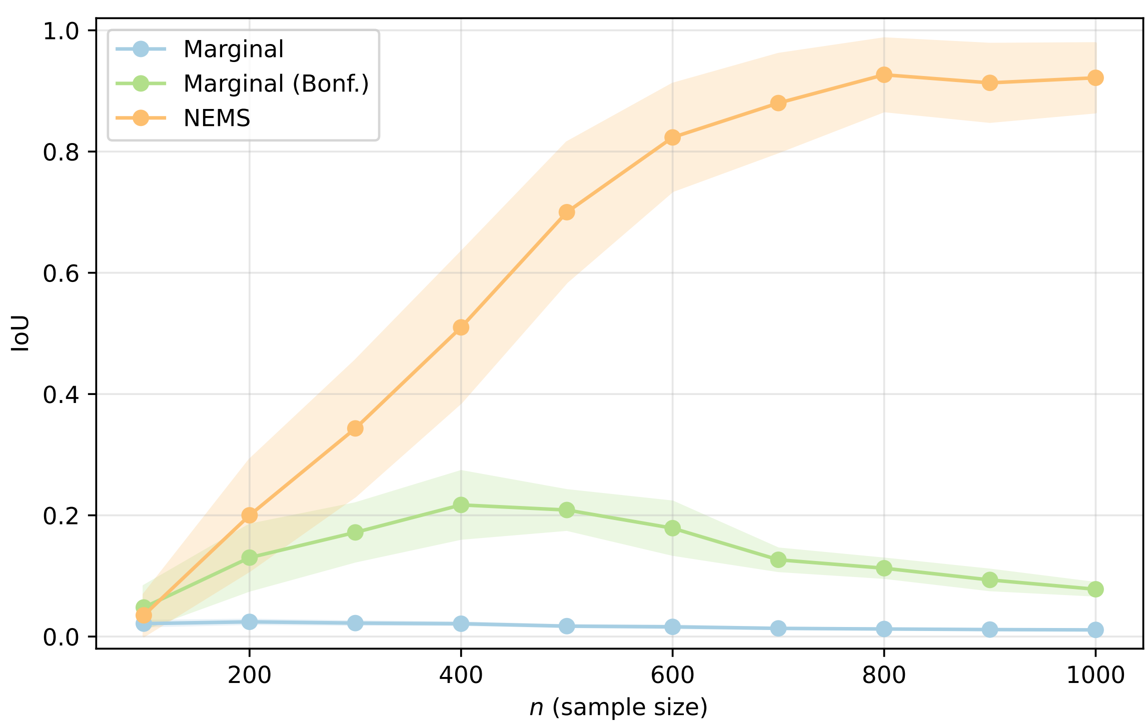

Figure 4: Synthetic experiments with quadratic heterogeneity, varying sample size, reporting IoU with 95% confidence intervals from 50 repetitions per configuration.

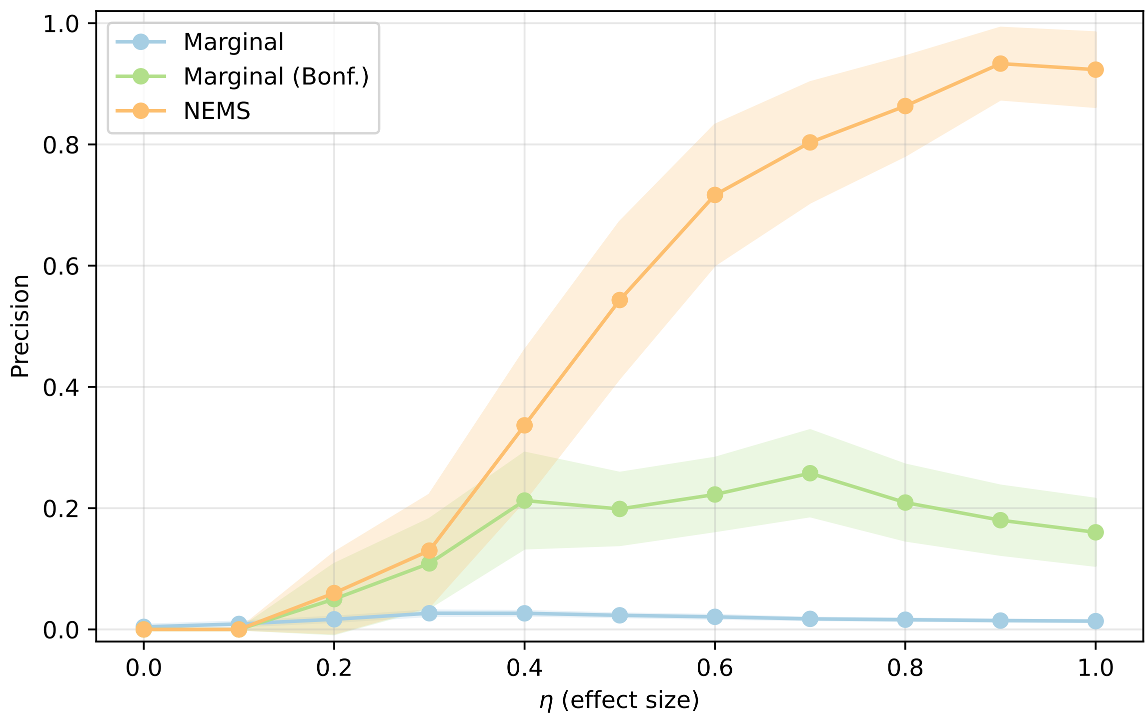

Figure 5: Synthetic experiments with quadratic heterogeneity, varying effect size, reporting precision with 95% confidence intervals from 50 repetitions per configuration.

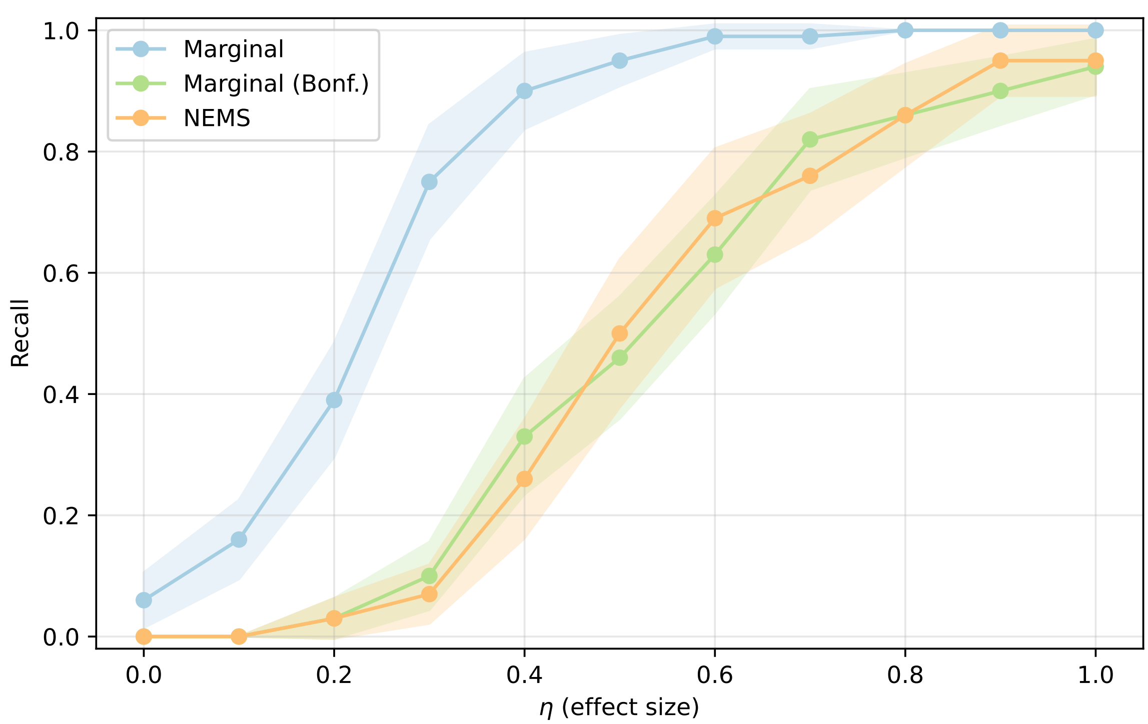

Figure 6: Synthetic experiments with quadratic heterogeneity, varying effect size, reporting recall with 95% confidence intervals from 50 repetitions per configuration.

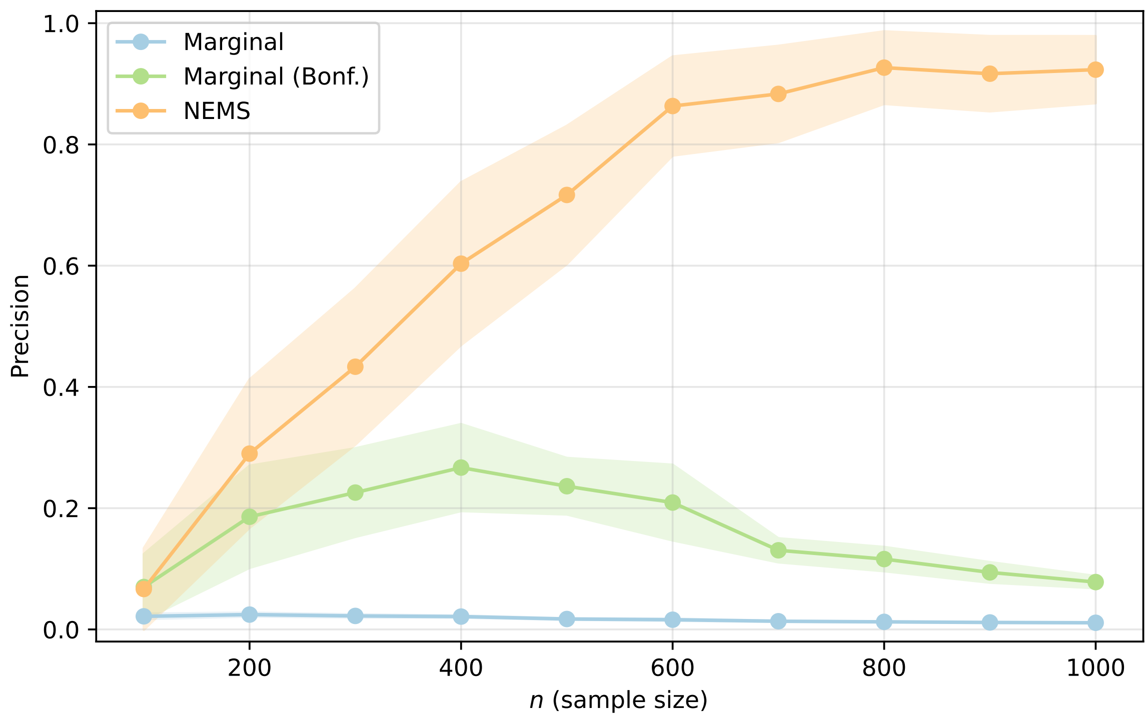

Figure 7: Synthetic experiments with quadratic heterogeneity, varying sample size, reporting precision with 95% confidence intervals from 50 repetitions per configuration.

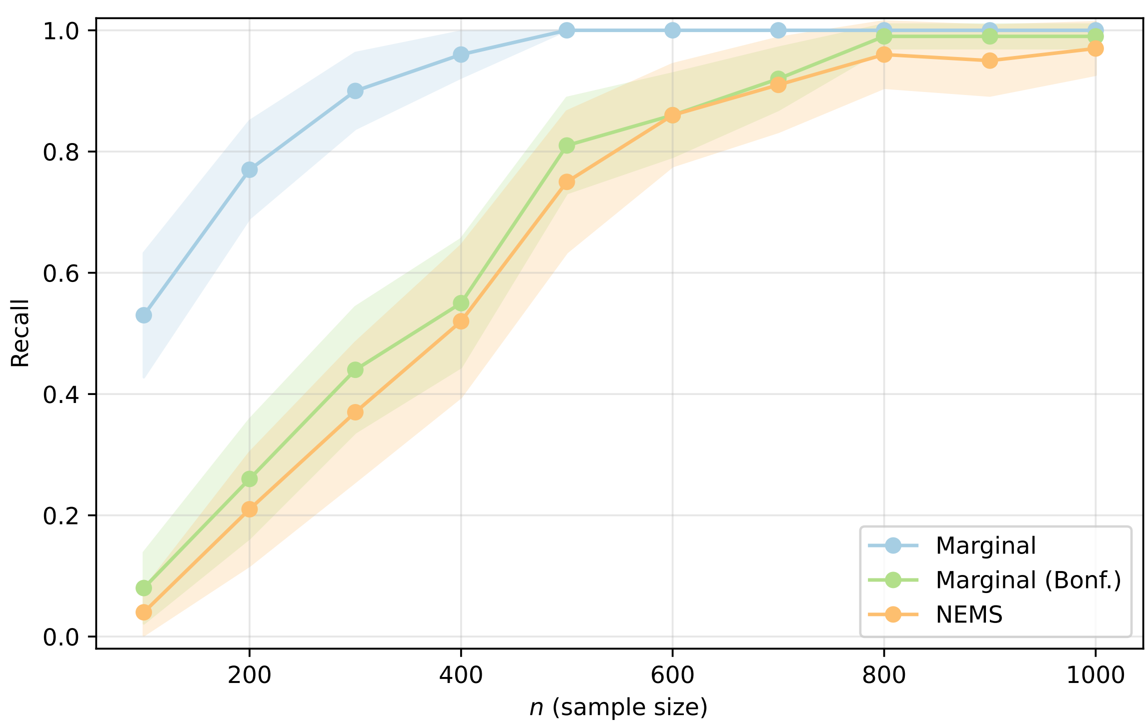

Figure 8: Synthetic experiments with quadratic heterogeneity, varying sample size, reporting recall with 95% confidence intervals from 50 repetitions per configuration.

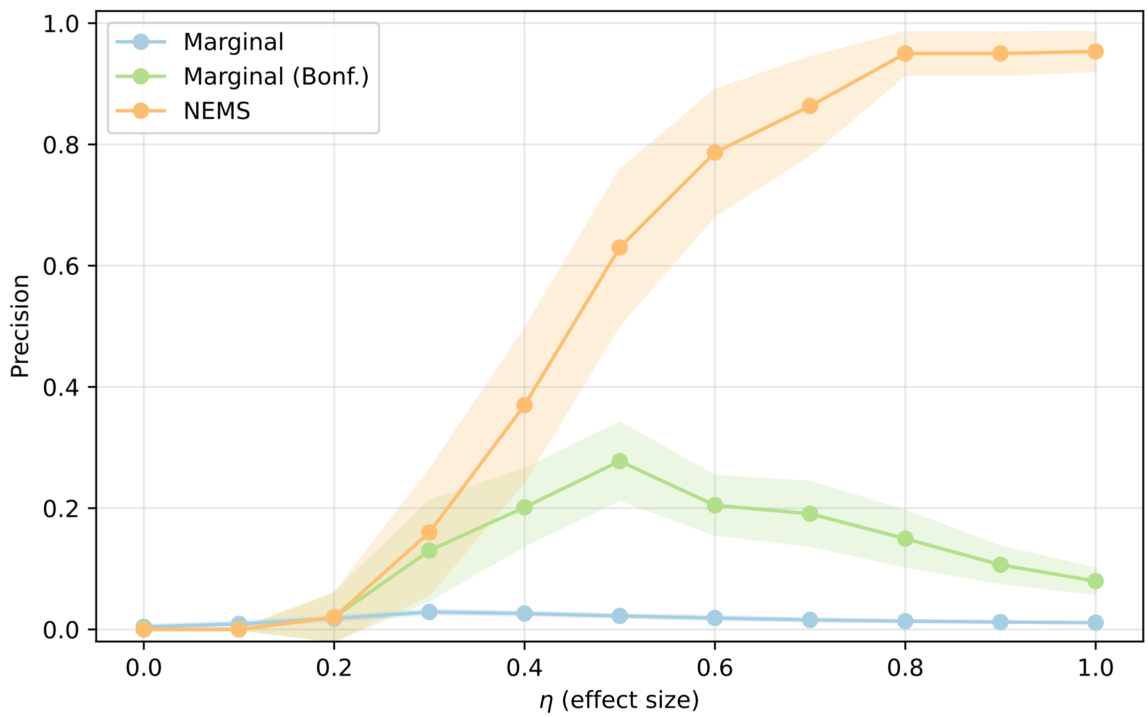

Figure 9: Synthetic experiments with linear heterogeneity, varying effect size, reporting precision with 95% confidence intervals from 50 repetitions per configuration.

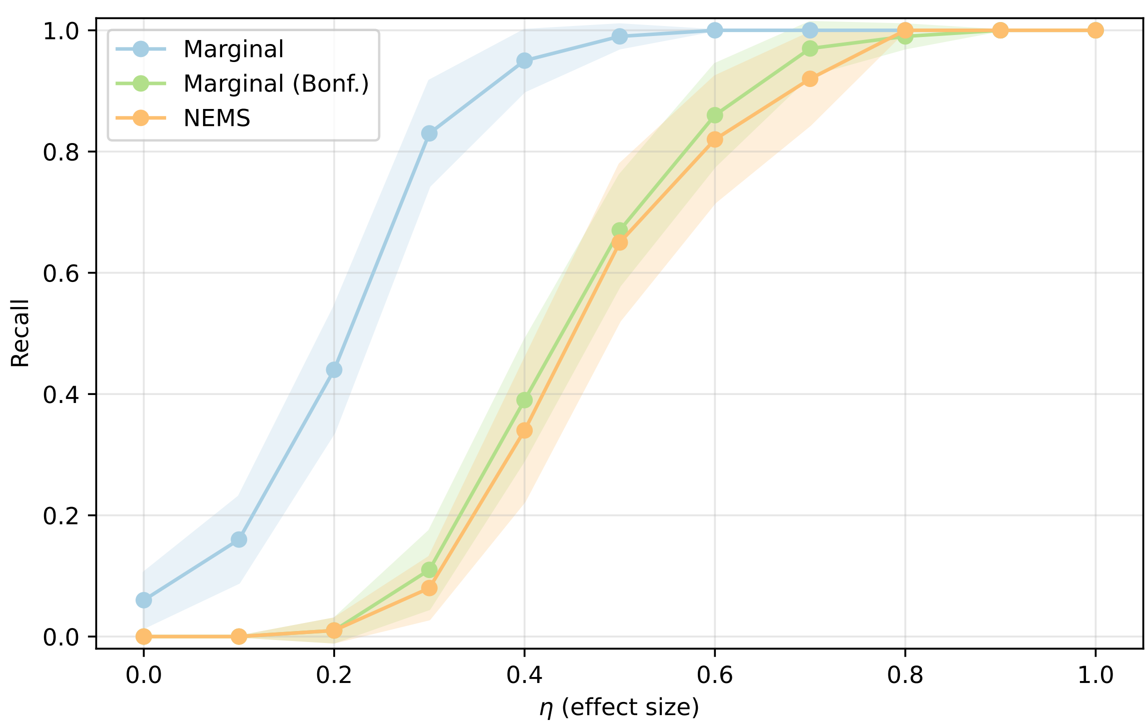

Figure 10: Synthetic experiments with linear heterogeneity, varying effect size, reporting recall with 95% confidence intervals from 50 repetitions per configuration.

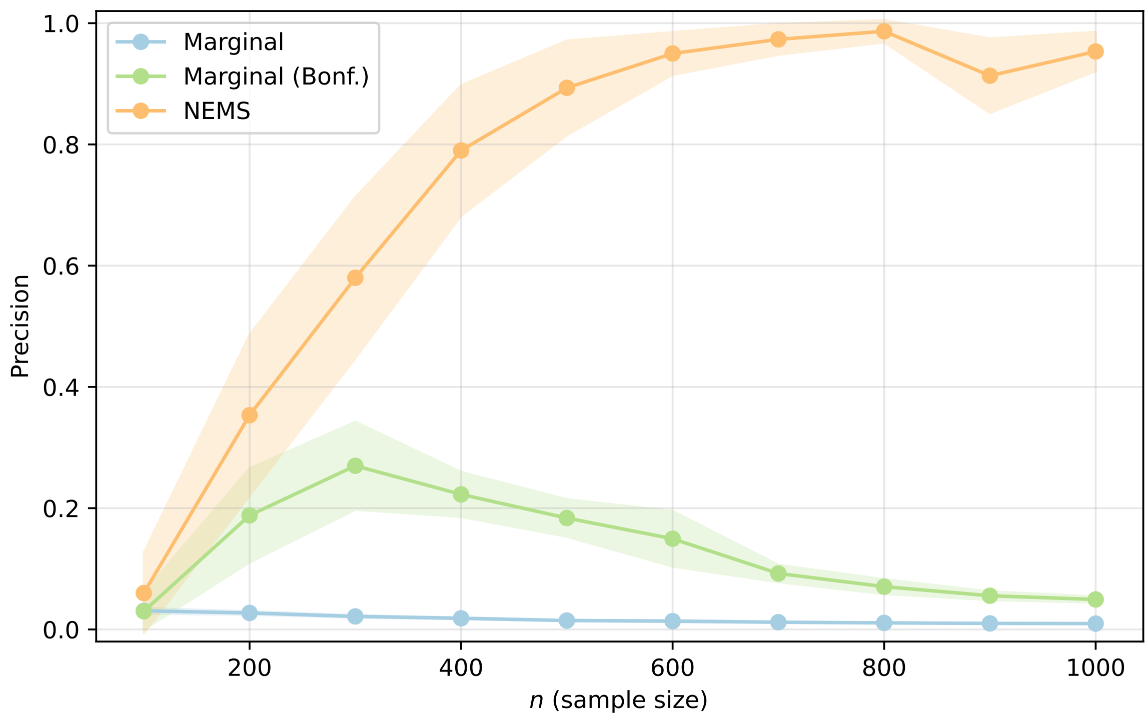

Figure 11: Synthetic experiments with linear heterogeneity, varying sample size, reporting precision with 95% confidence intervals from 50 repetitions per configuration.

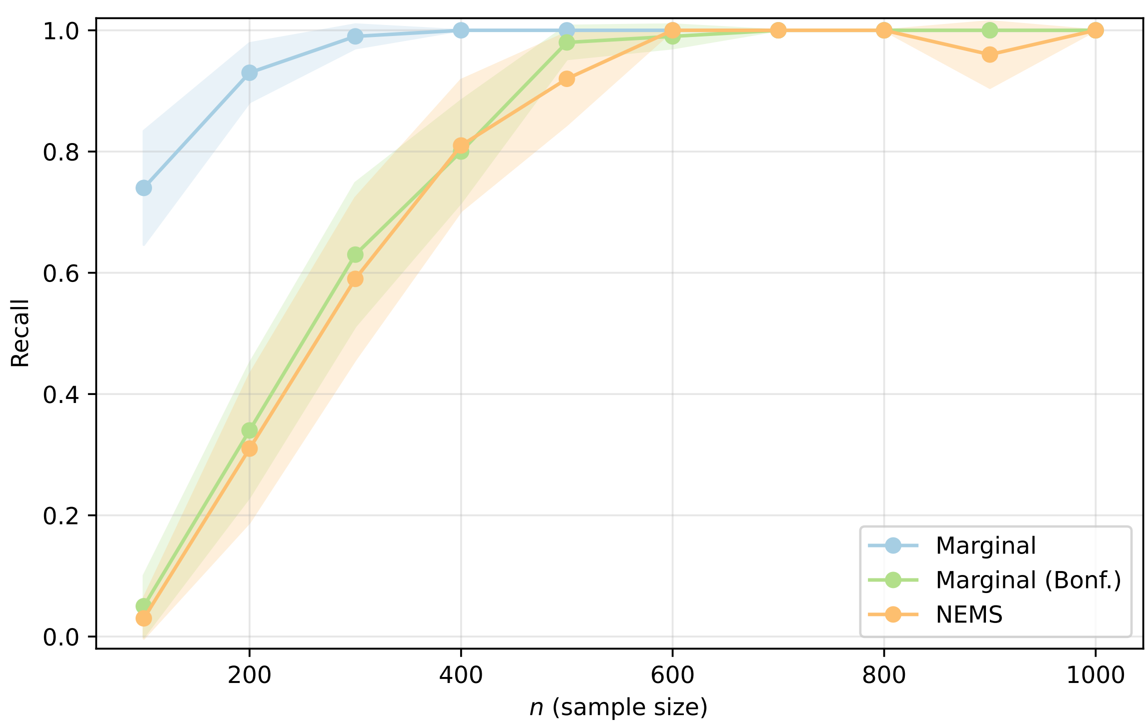

Figure 12: Synthetic experiments with linear heterogeneity, varying sample size, reporting recall with 95% confidence intervals from 50 repetitions per configuration.